

## Code-Layout, Readability and Reusability

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 18 July 2026                               |
| <b>Team ID</b>       | TBD                                        |
| <b>Project Name</b>  | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 5 Marks                                    |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                      |
|------|---------------------------|-----------------------------------------------|-------------------------------|------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | Proper indentation used.     |
| 2    | Proper File Structure     | Files and folders are logically organized     | yes                           | Organized project folders.   |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | yes                           | Meaningful variable names.   |
| 4    | Function / Method Names   | Functions are descriptively named             | yes                           | Clear function names.        |
| 5    | Code Comments             | Inline and block comments explain log         | yes                           | Comments added where needed. |
| 6    | Modular Design            | Code is split into reusable functions/m       | yes                           | Modular code structure.      |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | yes                           | No duplicate code.           |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | yes                           | Proper exception handling.   |

**Reusable Components / Modules:**

| S.No | Component / Module Name | Language / Technolog | Where Reused | Reusability Level (High / Medium / Low) |
|------|-------------------------|----------------------|--------------|-----------------------------------------|
| 1    | Authentication Module   | Python               | Login System | High                                    |

|   |                   |        |                   |        |
|---|-------------------|--------|-------------------|--------|
| 2 | Whisper Module    | Python | Speech-to-Text    | High   |
| 3 | Evaluation Module | Python | Similarity Score  | High   |
| 4 | Gemini Feedback   | Python | AI Feedback       | High   |
| 5 | PDF Report Module | Python | Report Generation | Medium |

**Overall Code Quality Assessment:**

| Aspect                   | Rating (1-5) | Comments                    |
|--------------------------|--------------|-----------------------------|
| Code Layout & Structure  | 5            | Well organized codebase.    |
| Readability              | 5            | Easy to understand.         |
| Reusability              | 5            | Modules reused effectively. |
| Documentation / Comments | 4/5          | Adequate comments provided. |
| Overall Score            | 4.8/5        | Excellent code quality.     |